

Lecture 6

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1 Standard Array

Let C be a linear $[n, k, d]$ code over $\mathcal{F} = GF(q)$. A standard array for C is a $q^{n-k} \times q^k$ array of elements of $\mathcal{F}^n$ defined as follows.

- The first row in the array consists of the codewords of C , starting with the all-zero codeword.
- Each subsequent row starts with a word $\mathbf{e} \in \mathcal{F}^n$ of the smallest Hamming weight that has not yet appeared in previous rows, followed by the words $\mathbf{e} + \mathbf{c}$, where $\mathbf{c}$ ranges over all the nonzero codewords in C in their order of appearance in the first row.

Claim: For any $[n, k, d]$ linear codes and $t = \lfloor \frac{d-1}{2} \rfloor$, all n -tuples of weight $w \leq t$ are coset leaders in a standard array.

Proof: We need to show that for all $x, y \in C$ having $w(x), w(y) \leq t$, they cannot be in the same coset. Let's assume to the contrary that x and y are in the same coset. Then

$$x - y \in C \implies wt(x - y) \geq d$$

However,

$$wt(x - y) \leq w(x) + w(y) \leq 2t < d$$

Contradiction. Hence such x, y always belong to different cosets.

By construction, the coset leader is the minimum-weight word in the entire coset, and thus all these x, y must be the coset leaders. Note that when the Hamming weight of the coset leader does not exceed $\lfloor \frac{d-1}{2} \rfloor$, then it is the unique minimum-weight word in its coset. This must be so in view of Maximum Error detection capability, where we have shown that nearest codeword decoding yields the correct codeword when the number of errors does not exceed $\lfloor \frac{d-1}{2} \rfloor$.

2 Probabilities of Correction/Detection

2.1 Error Detection

For a $[n, k, d]$ code, an error cannot be detected when the codeword is received, c_2 after combination of error pattern, e on the transmitted code c_1 also gives a valid codeword.

$$\implies c_2 - c_1 = e \implies e \in C$$

so, there are q^k possibilities for error patterns that cannot be detected over a q -ary channel.
 $\implies$ error patterns that can be detected $= q^n - q^k$.

For a $[n, k, d]$ code over a Binary Symmetric Channel with error probability p :

$$\mathbb{P}(\text{error detection}) = 1 - \sum_i^n A_i p^i (1-p)^{n-i}$$

where, A_i is the number of code words in C , with $\text{wt}(C) = i$.

$A_0 = 1$ (all zero $\in C$) and A_1 to $A_{d-1} = 0$ (minimum hamming weight = d).

2.2 Error Correction

$$\mathbb{P}(\text{error correction via coset decomposition}) = \sum_{i=0}^n \alpha_i p^i (1-p)^{n-i}$$

where α_i = number of coset leaders with weight i

3 Syndrome Decoding

For a $[n, k, d]$ code, the coset decomposition is $q^{n-k} \times q^k$ and it is quite difficult to store the entire coset decomposition, especially for large n and k .

Instead, we store the *syndrome* of the codeword which is defined as $s = xH^T$, where H is the parity-check matrix.

- Note that for every vector $\mathbf{c} \in \mathcal{F}^n$,

$$\mathbf{c} \in C \implies \mathbf{s} = \mathbf{c}H^T = \mathbf{0}$$

- All the vectors in any coset will have the same syndrome For example: Let $\mathbf{x}$ belong to a coset with leader e_l . Then

$$\begin{aligned} s = xH^T &= (e_l + c)H^T \\ &= e_l H^T + cH^T \\ &= e_l H^T \end{aligned}$$

- Different cosets will have different syndromes.

Proof: Let us assume the contrary:

For some l, j such that $l > j$

Assume $e_l H^T = e_j H^T$

$$\implies (e_l - e_j) H^T = \mathbf{0}$$

$$\implies e_l - e_j \in C \implies e_l = e_j + c$$

But e_l is the leader of a coset which comes after the coset with leader e_j .

But the above implies e_l is in coset j . This is a contradiction.

So, instead of storing the entire standard array, we just need to store the syndrome of all coset leaders in a table & match our received codewords syndrome with the table.

The procedure is:

1. Calculate $s = rH^T$
2. Use the table to find coset leader e_l corresponding to the syndrome s .
3. $\hat{c} = r - e_l$.

Example:

Consider a (5,2,3) code:

$$C = \{00000, 10111, 01101, 11010\}$$

The parity check matrix will be:

$$H = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$H^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The table for syndrome decoding in this example is:

e	syndrome
00000	000
00001	001
00010	010
00100	100
01000	101
10000	111
00011	011
00110	110

1. Suppose $C = 11010$ and error $e = 01000$ which is a coset leader,

$$r = c + e = 10010$$

$$s = rH^T = 101$$

which corresponds to $e = 01000$

$$\Rightarrow \hat{c} = r \oplus e_l = 10010 \oplus 01000 = 11010$$

Successfully decoded.

2. Now, assume $c = 11010$ but error $e = 10100$ which is not a coset leader,

$$\begin{aligned} r &= 01110 \\ B &= rH^T = 011 \\ \Rightarrow e_l &= 00011 \\ \hat{c} &= r \oplus e_l = 01110 \oplus 00011 \\ 01101 &\neq c \end{aligned}$$

So doesn't work if the error is not coset leader.

4 Important Linear Codes

4.1 Simple Parity Check

Simple Parity Check: $(k+1, k, 2)$

$$\begin{aligned} G &= [I_K \mid A] \\ A^T &= \underbrace{\begin{bmatrix} 1 & \dots & 1 \end{bmatrix}}_K \end{aligned}$$

Dual of this is, i.e. the parity check matrix is the generator for repetition code.

$$H = \underbrace{\begin{bmatrix} 1 & \dots & \dots \end{bmatrix}}_{K+1}$$

4.2 Hamming code

Hamming codes are also a type of linear block codes. They can detect up to 2 errors and, correct up to 1 error. A general Hamming code is characterized by a m such that $\forall m \geq 1$, we can make a $(2^m - 1, 2^m - m - 1, 3)$ Hamming code over $\text{GF}(2)$. Size of the parity check matrix $H = m \times (2^m - 1)$. It is generated by taking all the non-zero of length m and putting them as the columns in H .

Notice that the number of m -length non-zero binary vectors are $2^m - 1$ which is equal to the number of columns of H

For example for $m = 3$:

$$\begin{aligned} H &= \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -A^T & I_3 \end{bmatrix} \\ G &= \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} I_4 & A \end{bmatrix} \end{aligned}$$

Claim: Minimum distance $d = 3$ in a Hamming Code.

Proof: 1) No two-column vectors are linearly dependent

If they were to be linearly dependent, then we can find 2 column vectors v_1, v_2 such that $v_1 \oplus v_2 = 0$. But that would imply $v_1 = v_2$ which is not possible. Here

2) Can always find 3 vectors that are linearly dependent

For example, choose $v_1, v_2 \in$ columns of H . Then $v_3 = v_1 \oplus v_2 \in H$ as $v_1, v_2 \neq \mathbf{0}$. This follows by the general construction of Hamming Codes. Thus we can find $v_1, v_2, v_3 \in H$ such that $v_1 \oplus v_2 \oplus v_3 = 0$.

Thus minimum distance in a Hamming Code is 3.

5 Perfect Codes

Let $[n, k, d]$ be a code over the alphabet $\mathcal{F}$ of size q . Any such code satisfies the **Hamming Inequality** given by:

$$|C| \leq \frac{q^n}{\sum_{i=0}^t \binom{n}{i} (q-1)^i}$$

where $|C|$ = number of code words and $t = \lfloor \frac{d-1}{2} \rfloor$

Any code that satisfies the above inequality with equality is called a **Perfect Code**.

For binary codes over $GF(2)$, $q = 2$ thus the inequality reduces to

$$|C| \leq \frac{2^n}{\sum_{i=0}^t \binom{n}{i}}$$

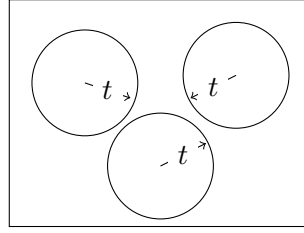
Intuition: Total number of words in $\mathcal{F}^n$ is given by q^n

Further, a sphere of radius t in $\mathcal{F}^n$ is a set of words in $\mathcal{F}^n$ at Hamming distance t or less from a given word in $\mathcal{F}^n$. The number of words in such a sphere, or the volume of the sphere, is given by

$$V_q(n, t) = \sum_{i=0}^t \binom{n}{i} (q-1)^i$$

Each codeword $c \in C$ creates a Hamming ball around itself of radius t which is non-intersecting in nature (otherwise any-word in the intersection would be impossible to decode correctly) The total number of words contained in these Hamming Balls are given by $|C|V_q(n, t)$ and this must be less than or equal to the total possible number of words q^n . Thus

$$|C|V_q(n, t) \leq q^n \implies |C| \leq \frac{q^n}{\sum_{i=0}^t \binom{n}{i} (q-1)^i}$$



Equality occurs when every codeword in $\mathcal{F}^n$ is included in some or the other Hamming Ball.

Claim: Hamming Codes are Perfect Codes

Proof: A general Hamming code is given by $[2^m - 1, 2^m - m - 1, 3]$

Here $n = 2^m - 1$ and $k = 2^m - m - 1$. $t = \lfloor \frac{d-1}{2} \rfloor = 1$.

Total length of Codeword = $n = 2^m - 1$

Number of Coset leaders with weight = 2^m . This includes the $\mathbf{0}$ vector and the $2^m - 1$ vectors with weight 1.

$|C| = \text{number of codewords} = q^k = 2^k$

Thus

$$|C| = 2^k \leq \frac{2^n}{2^m}$$

which is satisfied with equality as $n = k + m$.

Perfect Codes in general are quite rare. The (23,12,7) Golay Code is the only other Perfect Code besides the Hamming Code over $GF(2)$.

6 General Q-ary Hamming Code

A general q -ary Hamming code over the field $GF(q)$ can be represented as

$$\left[\frac{q^m - 1}{q - 1}, \frac{q^m - 1}{q - 1} - m - 1, 3 \right]$$

Its parity check matrix H is generated by taking all non-zero vectors of length m which have 1 as their first non-zero component. As every pair of columns is linearly independent, $d \geq 2$. In fact, we can find three columns that add up to give $\bar{0} \implies d = 3$

Example: for $m = 3$ over field $GF(3)$

$$H = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 0 & 0 & 1 & 1 & 0 & 1 & 2 & 2 & 0 & 2 & 1 & 2 \end{bmatrix}$$

7 Decoding Hamming Codes

For a $[7, 4, 3]$ systematic Hamming Code

$$c = (\mu_0 \quad \mu_1 \quad \mu_2 \quad \mu_3 \quad p_0 \quad p_1 \quad p_2)$$

$$p_0 = \mu_0 + \mu_2 + \mu_3$$

$$p_1 = \mu_0 + \mu_1 + \mu_2$$

$$p_2 = \mu_1 + \mu_2 + \mu_3$$

- Now syndrome $s = Hc^T$

$$\begin{bmatrix} s_0 \\ s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mu'_0 \\ \mu'_1 \\ \mu'_2 \\ \mu'_3 \\ p'_0 \\ p'_1 \\ p'_2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} s_0 \\ s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} p'_0 + \mu'_0 + \mu'_2 + \mu'_3 \\ p'_1 + \mu'_0 + \mu'_1 + \mu'_2 \\ p'_2 + \mu'_1 + \mu'_2 + \mu'_3 \end{bmatrix}$$

Which is basically checking the parity bits and whether the parity is justified or not.

- Another option is to just find the column of H that matches with the syndrome and flip that bit. Since

$$r = c + e \Rightarrow s = Hr^T = Hc^T + He^T = He^T$$

Since $w(e) = 1 \Rightarrow He^T$ is simply the index of the column which comes out as s .

For example if $e = (0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)$

$$s = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

The syndrome matches with the 4th column of $H \Rightarrow$ we flip the 4th bit in r to recover the message. This however requires for us to iterate over all the columns of H and check with which column it matches up.

- If we represent H in a particular way by taking the columns of H in order of their increasing binary representation, we don't need to match the syndrome with the columns of H . We just need to flip the bit that corresponds to the binary number of the syndrome.

For example, say $C = 1010101$ and $r = 1000101 \implies e = 0010000$. $s = Hr^T = He^T$

$$s = He^T = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

As $s = [0 \ 1 \ 1]$ corresponds to the decimal number 3 $\implies$ flip the 3rd bit in r to recover the message the original message $C = 1010101$.

8 Extended Hamming Code

We can add a total parity check bit to a Hamming code to increase its minimum distance. The general form of hamming code over binary field is $[2^m - 1, 2^m - m - 1, 3]$, adding a total parity check bit results in $[2^m, 2^m - m - 1, 4]$.

Claim: The minimum distance increases to 4.

Proof: Let us consider a pair of codes $p, q \in$ ordinary Hamming Code.

- Case 1: The Hamming distance between p and q is greater than 3. As adding an extra bit cannot decrease the distance between two codewords, the Hamming distance between any two such words after adding a parity check bit is still greater than 3.
- Case 2: The Hamming distance between p and q is 3. This implies that the total parity of one of the codes is even and it is odd for the other. The total parity check bit will hence be different. Hence the Hamming distance for such a pair will now be 4.

Hence we can conclude the minimum distance for the extended Hamming code is 4.

The extended Hamming code can decode up to 3 errors which can be seen from the formula:

$$2t + l \leq d - 1$$

The Parity check matrix for this code is:

$$H = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$